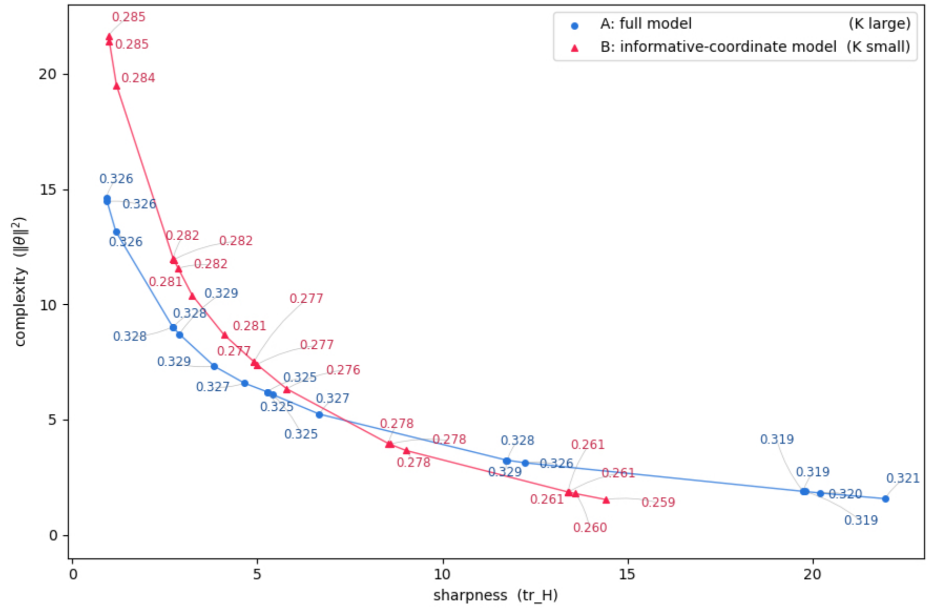
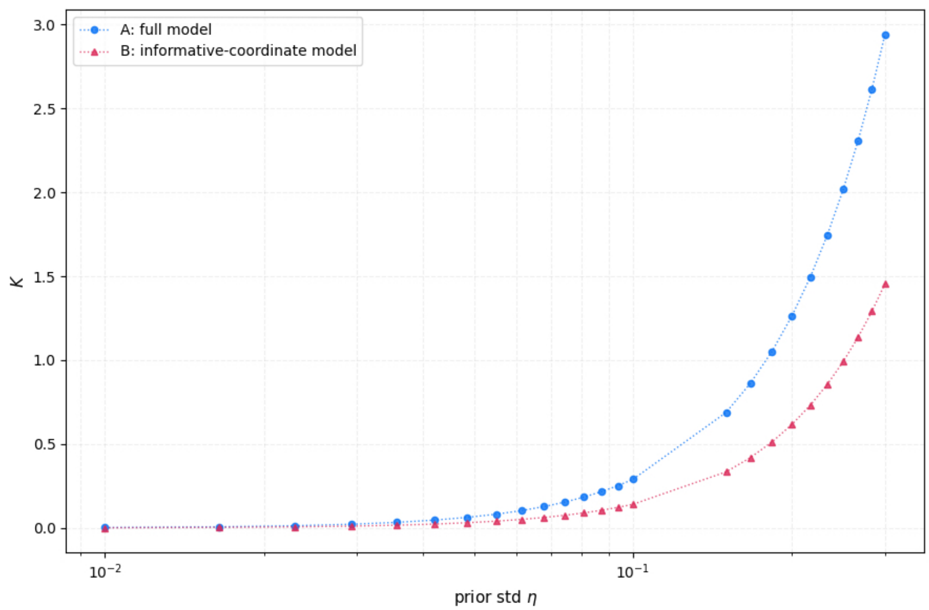


(a) Empirical Pareto frontiers (S, KL) of the two simple models. Each point represents a hyperparameter setting (SGD learning rate and weight decay). As the problem is convex, all points lie exactly on the frontier. The text annotations denote the *generalization gap* (test loss − training loss), *smaller is better*, training loss for all trained models is very close to 0.



(b) K values of the two simple models under prior $\mathcal{N}(0, \eta^2 \cdot I)$, $\eta = \{0.01, \cdots, 0.30\}$. By owning more prior information, Model B has a significantly smaller K.

Figure 1: **Simple Controlled Study. Setup:** Logistic regression task (model parameter dimension $d = 200$, training data size $n = 90$) where only the first 100 coordinates of the ground-truth $\theta^\star$ are informative ($\theta^\star_{1:100} \sim \mathcal{N}(0,1)$, $\theta^\star_{101:200} = 0$). Model A *(blue)*: all 200 coordinates are learnable; Model B *(red)*: only learns the 100 informative coordinates, and the rest are fixed to 0. **Takeaway:** If (S, KL) alone could explain generalization, Model B should perform worse in the left region of the plot, as it has larger S at the same KL. However, the annotations show Model B consistently achieves *smaller* generalization gap. This contradiction is resolved when incorporating K: Model B has a smaller K. This supports our claim that (S, KL) alone is insufficient, and K is essential to fully explain generalization.